Speaking Script — Slides 7 and 8

Everything on each slide, every chart explained, why each result happened, and the mathematics behind the PINN. Normal text is what you say out loud. Green is background you should understand but need not read aloud. Red is a warning. Speaking time: about 6 minutes for both slides.

SLIDE 7 — Where we were before physics

What is on this slide

Where	What it is
Top left	Title and one-line subtitle
Big chart, left	Six bars, one per method we tried. Below zero = it made things worse
Caption under chart	Two lines summarising the failures
Three cards, right	Six attempts none worked / model barely beat guessing / what we concluded
Three number boxes	6 methods tried · 0 that worked · 53 out of 100
Grey bar, bottom	The sentence that sets up slide 8

The opening

"Before I show you the physics result, I need to show you where we were. This slide is every method we tried before it."

"The rule for every experiment was the same. Build the model with the new idea in it. Then build an **identical** model with that idea switched off. Same size, same training, same random seeds. Compare the two. That way any difference is caused by the new idea and nothing else."

Explaining the chart, bar by bar

"The black line across the middle is the control — the model without the new idea. Anything **above** the line means the idea helped. Anything **below** means it hurt. Every single bar is below the line."

"Bar one, **reweighting**, minus eleven. This is the standard first thing anyone tries. You tell the model to care more about quiet areas. It made the gap **worse**."

"Bars two and three, **Group-DRO** and **adaptive thresholds**. Almost exactly zero. We tried three different strength settings for Group-DRO. Nothing moved at all."

"Bar four, the **plain graph network**, minus six point five. This let neighbouring areas share information. It actively damaged the model."

"Bars five and six, **gated and attention graphs**. These are the clever versions, specifically designed to fix the problem the plain one has. About minus one each. Not damaging, but not helping either."

Why each one failed — if he asks

Method	Why it failed
Reweighting	Telling the model to care more about quiet areas makes it shout "crime" there more often. It catches a few more real crimes but produces many more false alarms, so the score falls. You cannot create knowledge by shouting louder.
Group-DRO	It optimises for whichever group is scoring worst. But the worst group scores badly because crime is **rare** there, not because the model is ignoring it. Focusing harder on a group does not change how rare its events are.
Adaptive thresholds	Moving the decision cut-off only trades precision against recall. It slides you along a curve; it does not lift the curve. No new information enters the model.
Plain graph (GCN)	Every layer averages an area with its neighbours, and it cannot stop. Like mixing paint — after three or four layers every area is the same muddy colour and the model can no longer tell them apart. This is called over-smoothing.
Gated / attention graph	These have a switch that decides how much neighbour information to take in. What they learned was to turn it almost **off**. So they behave like the no-graph model, minus a small cost for carrying machinery they do not use.

The three cards on the right

"Six attempts, none worked. Not one added ingredient beat its own matched control."

"And the level we were stuck at was low. On the burglary data the model scored **53 out of 100** at telling a crime week from a quiet one. Fifty is a coin flip. So it was barely better than guessing."

"Our conclusion at that point was that the limit was in the **data**, not in the architecture. More layers, better graphs, smarter losses — everything hit the same wall."

Closing this slide

"So that is the honest starting position. Six methods, zero successes, and a model barely above chance. **This is why the next slide matters** — the physics is the first ingredient in the whole project that earned its place."

Saying the failures plainly is a strength, not a weakness. It makes the next slide credible. If you skip straight to the good result he will reasonably ask "compared to what?"

SLIDE 8 — What changed with physics

What is on this slide

Where	What it is
Big chart, left	Three bars: generic smoothing 44.6, no physics 53.2, with physics 71.5. Dashed line at 50 = pure guessing
Caption under it	5 seeds, t = 9.6, and the ablation result
Two small charts, right	Left panel: skill in poor areas. Right panel: how often it cries "crime"
Four number boxes	53→71 · 51→58 · 3.2x→1.8x · minus 23%
Yellow bar, bottom	The honest limitation

The opening — explain the rule in plain words

"Then we tried something different. Instead of changing the network shape, we wrote a **real criminology rule** into the model."

"The rule is this. When a house gets burgled, the street around it becomes more attractive to burglars for a while. Criminologists call this repeat and near-repeat victimisation, and it is measured, not invented. That extra attractiveness spreads to nearby houses, and it fades away over time."

"Short and Bertozzi turned that idea into two equations in 2008. We wrote those equations into the training so the model is not only fitting numbers — it also has to obey how burglary is known to behave."

The main chart

"The dashed line at fifty is pure guessing. The grey bar is our old model at **53** — barely above it. The green bar is with physics: **71.5**. That is a jump of eighteen points."

"Five separate training runs each. The t-value is nine point six, which means this is not luck."

"Now the red bar on the left is the important one, and I want to be clear about why it is there."

"Someone could say: maybe the equations are not doing anything clever, maybe any smoothing rule would help. So we tested exactly that. We replaced the burglary equations with a **generic smoothness rule** that contains no crime knowledge at all. It scored **44.6** — **worse than having no rule**, and below the guessing line."

"So it is not smoothing. It is specifically the burglary equations doing the work."

This is the single most important defensive point on the slide. It is the same control logic as the rest of the project: compare against the thing that isolates the mechanism, not just against nothing.

The two small charts

"On the left, skill in the poorest neighbourhoods. It went from **51** — which is a coin flip — to **58**. So the physics helps **most** where the data is thinnest. That makes sense: when you have very little data, having a rule helps more."

"On the right, false alarms. In poor areas a burglary really happens in about **28.5%** of weeks — that is the red dashed line. Our old model shouted "crime here" in **90%** of weeks. Three times too often. With physics and the corrected loss it shouts in **52%** of weeks. Still too many, but nearly halved."

"That matters practically. If a system flags a poor neighbourhood every single week, police keep being sent there whether or not anything is happening."

The honest limitation — say this yourself

"One limitation I want to state clearly. The model learned **where** burglary happens. It did not learn **when**. Within any one neighbourhood it is still close to guessing week to week."

"It is like knowing it rains a lot in London. True and useful — but it does not tell you about next Tuesday."

Volunteering this is much better than him finding it. It also protects you: if a reviewer later notices the per-group scores are near chance, you already said so.

The mathematics — for when he asks

He will ask. This section is what you say.

1. What a PINN is, in one sentence

"A normal network learns only from data. A physics-informed network learns from data **plus a known equation**. The equation is written into the loss, so the network is punished both for getting the data wrong **and** for breaking the physics."

2. The two quantities the model tracks

Symbol	Meaning
A(x, y, t)	**Attractiveness** — how appealing a location is to burgle, at that place and time
rho(x, y, t)	**Offender density** — how many potential offenders are around there
rho × A	**The crime rate** — this is the only thing we actually observe in the data

3. The two equations (Short et al. 2008)

These are the equations we wrote into the model:

```
dA/dt = eta * Lap(A) - A + A0(x,y) + rho*A

drho/dt = Lap(rho) - 2*div( (rho/A) grad A ) - rho*A + A - A0(x,y)
```

"Lap means the Laplacian — it measures how much a value differs from its surroundings, so it describes spreading. Grad means the gradient — which direction things increase in."

4. What every single term means

Term	In plain words
eta * Lap(A)	Attractiveness **spreads** to nearby locations. eta controls how fast.
- A	Attractiveness **fades** over time. A burgled street does not stay hot forever.
+ A0(x, y)	Every place has a **baseline** appeal that does not change — wealth, layout, how easy it is to escape. We let the model learn this as a map.
+ rho*A	**A crime raises attractiveness.** This is repeat victimisation — the heart of the model.
Lap(rho)	Offenders **wander** randomly.
- 2*div((rho/A) grad A)	Offenders **drift towards** attractive places. They move up the attractiveness hill.
- rho*A	After committing a crime an offender **leaves**.
+ A - A0	**New offenders appear** to replace them.

5. How the equation gets into the training

The total loss has two parts:

```
Loss = Loss_data + lambda * Loss_physics
```

"**Loss_data** says: the predicted crime rate should match the burglaries we actually recorded."

"**Loss_physics** says: whatever the network outputs must satisfy those two equations. We take the network output, plug it into the equation, and see what is left over. That leftover is called the **residual**.

If the physics holds, the residual is zero. We square it and add it to the loss."

```
Loss_physics = mean( residual_A^2 ) + mean( residual_rho^2 )
```

"We check this at ten thousand random points across the city and across time, each training step."

The derivatives are computed by automatic differentiation, not by finite differences. The network is a smooth function of x, y and t, so PyTorch can differentiate it exactly. That is what makes a PINN possible.

6. Why we changed the loss to Poisson

"Burglaries per square per week are **counts** — 0, 1, 2, 3. Counts follow a Poisson distribution, not a bell curve."

"The usual squared-error loss treats them like a bell curve. That means busy squares, which have big numbers, dominate the training. The model is effectively tuned for rich areas."

"The Poisson loss weights each square correctly for its own rate. That is why false alarms in poor areas fell from three times reality to under two — not a trick, a correct assumption."

```
Poisson negative log-likelihood: predicted_rate - observed_count *
log(predicted_rate)
```

7. What the model discovered by itself

"We did not fix the constants. We let the model learn them from Chicago data."

"It learned a spreading rate **eta of about 0.012 to 0.017**, consistently across all five runs. And it learned A0 as a **map** of baseline appeal across the city, not one number."

A small eta means attractiveness spreads only a short distance. That is consistent with the criminology: near-repeat burglary effects are measured over metres and weeks, not kilometres and years. The model recovering that from data on its own is a genuinely good sign.

8. Why AUC and not F1 — the one-liner

"AUC asks: show the model one week with a burglary and one without — can it point to the right one? Fifty is a coin flip. It cannot be fooled by how rare crime is, and it cannot be fooled by moving the decision cut-off. F1 can be fooled by both. That is why we lead with AUC."

If you only remember four numbers

Number	What to say
53 → 71	"From barely-better-than-guessing to genuinely useful."
44.6	"Generic smoothing scored worse than nothing, so it is the burglary equations, not regularisation."
3.2x → 1.8x	"False alarms in poor neighbourhoods nearly halved."
t = 9.6	"Five seeds. This is not luck."

Three things not to say

1. Do not say the PINN "beat the GNN". Different datasets, different tasks. Say: the physics beat its own matched control, and the graph did not.

2. Do not claim the model predicts crime well. It predicts **where**, not **when**. Say that before he asks.

3. Do not skip the 44.6 bar. Without it the whole result can be dismissed as "you just added a smoothing penalty".